

WorldLearning: Visualizing How Local People Have Fun Across Countries Using Large Language Models

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Abstract—People around the world share broad experiential goals such as “relaxing” or “having fun,” but the concrete activities that realize these experiences vary widely by place. Existing tools like search engines, guidebooks, travel blogs, and large language models (LLMs) can answer questions such as “What do people do for fun in Japan?”, yet they typically present one country at a time and require users to already know what to ask. These answer-first interfaces make it hard to see how a single experience is realized differently across countries, or to surface surprising contrasts that might provoke deeper questions. This paper presents *WorldLearning*, an interactive system that uses an LLM and a world map to help users explore how local people in different countries typically have fun in their free time. Given a user-specified activity (for example, “going to karaoke”), *WorldLearning* uses an LLM as a scoring model to estimate how typical that activity is as something locals do for fun in each country, and visualizes the resulting scores as a choropleth map with ranking lists and follow-up explanation tools. The system is designed to transform interaction with LLMs from one-shot question-answer exchanges into a dialectical process in which visualized patterns generate new questions that users then probe further. A preliminary study with two participants engaging in guided think-aloud exploration suggests that *WorldLearning* can surface unexpected global patterns, prompt questions that users would not have asked on their own, and support sustained curiosity-driven inquiry about why everyday experiences are realized differently across countries.

I. INTRODUCTION

In this paper, I use *experience* to refer to broad intentions such as relaxing or having fun. The same experience can be realized differently across the globe. For example, people pursue the goal of “having fun” through different concrete activities in different places. In Japan, local people go to karaoke for fun. While in the US, people watch football for fun. In other words, the underlying experience is shared, but the specific activities that realize it vary by context. Yet when we try to learn about life in other countries, these everyday experiences are surprisingly hard to see. Students and curious learners are often presented with national statistics, famous landmarks, or broad cultural stereotypes, rather than concrete activities that local people actually do. Information about leisure tends to appear as scattered anecdotes or travel tips, not

as a structured view that lets us compare experiences across places. As a result, it is difficult to ask simple but meaningful questions like “Where is this activity really local?” or “In which countries do people actually do this for fun?”, and even harder to notice when the answers contradict our expectations.

There are some existing tools that already can answer the question of what people do for fun in other countries. If I am curious about one place, I can search “things to do in Tokyo,” read guidebooks and blogs, or watch vlogs, and I will get a reasonable picture of possible activities. I can also ask an LLM “What do people do for fun in Japan?” and receive a coherent description. But all of these tools share the same underlying assumption: the user already knows what to ask. People can only frame questions that fit their existing intuitions about a place. If they do not realize that a certain activity is especially common somewhere, or that two countries differ in a surprising way, they will never think to ask the question in the first place.

These tools fail to solve the problem of helping a user systematically explore how a single experience like “having fun” is realized differently across countries, in a way that provokes curiosity. Search engines, travel sites, and guidebooks are answer-first and place-by-place: they give lists for one country at a time, but they do not let the user see, in one view, how realizations of “having fun” line up across many countries or where an activity is central versus marginal. Because they never put these realizations side by side, they also do not surface interesting patterns or unexpected differences that could become questions, such as “Why is this activity common here but rare there?” LLMs encode a rich internal model of “what people do,” yet that structure is hidden behind one-off text answers. Users never see how strongly the model associates karaoke with Japan compared to other activities or other countries, and once they receive a paragraph, the interaction usually stops, and they do not encourage users to probe deeper into specific experiential contrasts.

In this paper, I address this challenge with *WorldLearning*, an interactive system that uses a world map and a large language model (LLM) to help users explore how a broad

experience, such as “having fun,” is realized through concrete activities across countries. At a high level, WorldLearning lets a user choose an experience, input an activity, and then immediately see a world map of where that activity appears more or less typical in everyday local life. WorldLearning uses the language model as a scoring model for each country and visualizes the resulting scores as a choropleth map that users can hover over, compare, and click into. By presenting the whole pattern at once, the system is designed not only to inform but to surface unexpected differences that invite users to ask why an activity is common in some places and rare in others. The core idea is to externalize the model’s latent associations between activities and places so that users can see these patterns directly, rather than only receiving a single text answer to a single question. Figure 1 shows this main view.

WorldLearning is designed around the challenges and opportunities outlined above. Instead of assuming that users already know what to ask, the system surfaces patterns first: the map view reveals where an activity appears locally typical or atypical across countries, so users can notice contrasts they would not have thought to query and begin asking “why here and not there.” Underneath this interaction, WorldLearning uses standardized prompts and token log probabilities to derive relative plausibility scores that approximate how locally typical an activity is for a given experience in each country. Beyond the initial map, the system keeps users engaged and probing deeper by letting them inspect specific countries, compare multiple activities, and view language-model-generated explanations that connect visible differences to possible cultural, environmental, or institutional factors. The novelty is both in using an LLM as a local typicality model over countries and in organizing interaction around a curiosity loop, where visualizations help users generate and refine questions over time rather than simply providing one-shot answers.

I conducted a small study to see how WorldLearning supports curiosity-driven exploration in practice. Two participants used the interface in guided, think-aloud sessions starting from the prompt “What do local people do for fun?” I collected observational notes on what they noticed, the questions they asked, and the comparisons they made. Even with only two users, the sessions suggested that WorldLearning achieved its main goal: both participants discovered experiential contrasts they said they “would never have thought to look up,” treated the map as a trigger for new “why here and not there?” questions, moved back and forth between activities and countries, and used model-generated explanations as starting points for their own reasoning. While preliminary, these observations provide initial evidence that exposing an LLM’s experiential associations in map form can make local realizations of “having fun” visible and comparable, and help users generate and refine questions rather than simply retrieve answers.

This paper makes the following contributions: 1. I introduce **WorldLearning**, an interactive system that uses a world map and a large language model to visualize how broad experiences

such as “having fun” are realized through concrete activities across countries. The system externalizes the model’s associations between activities and places into a comparative view and structures interaction around noticing surprising patterns and then probing deeper. 2. Conceptually, this shifts how we interact with LLMs: instead of starting from a well framed question and expecting a single answer, WorldLearning treats the model as an inspectable prior about everyday life and uses visualization to create a dialectical interaction process, where the system first proposes patterns, the user responds with questions and hypotheses, and the system then provides further patterns and explanations that help sustain exploration and curiosity.

A. Related Work

WorldLearning connects to work on regional experience differences, experiential views of place, global map based exploration, and exploratory interaction with AI models.

Recent work has started to treat “what it is like” to visit a place as something that can vary systematically by region, even within the same category. The Yelp study on “Computationally Surfacing Regional Experience Differences from Location Based Services” holds a venue category fixed, such as bakeries or casinos, and uses text mining over reviews to test specific hypotheses about how the experience differs across states (for example, King Cake rituals in Louisiana versus Pennsylvania bakeries). By computing differential salience of phrases for each region–category pair, the authors recover many guidebook described contrasts and show that environmental, cultural, social, and institutional factors rewrite what a category affords in different places. That work demonstrates that regional experiential differences exist and can be surfaced computationally when one has large amounts of behavioral text. WorldLearning shares this focus on experience and difference, but at a different level. Rather than starting from a fixed set of hypothesized contrasts and a specific platform, it treats any user supplied activity as a probe into a model of local leisure across all countries, and surfaces where that activity appears typical or atypical as a starting point for curiosity.

HCI has proposed many systems that move beyond utility based recommendation toward what places feel like or afford. Examples include route planners that optimize for “beautiful” or “quiet” paths instead of only shortest time, systems that recommend venues based on ambiance descriptors such as “romantic,” “relaxing,” or “hipster,” and tools like Affinder that help users search for situations that afford particular activities in their city. These systems highlight the importance of experiential qualities and everyday practices, but they are usually grounded in a single city or region and focus on helping an individual choose where to go. WorldLearning is inspired by this experiential orientation, yet its goal is comparative and reflective rather than prescriptive. It does not tell a user where to go; instead, it reveals how a shared experience like “having fun” is realized through different activities across countries, and invites users to think about both difference and shared humanity.

There is a rich tradition of using thematic maps to help people see global patterns and challenge their assumptions. Work such as Gapminder’s animated visualizations shows how indicators like income and life expectancy evolve across countries, allowing viewers to discover clusters and outliers that contradict stereotypes. Systems like Atlasify and Cartograph use Wikipedia derived embeddings to project arbitrary concepts onto maps and let users explore semantic relatedness across regions. These tools demonstrate that map based overviews can powerfully support exploratory comparison. WorldLearning adopts similar visual language, a choropleth world map and ranking lists, but visualizes a different kind of quantity: the typicality of everyday activities under a language model’s beliefs, rather than measured statistics or document similarity. The contribution is not a new visualization technique but a new use of maps to externalize a model’s internal view of everyday life.

Finally, WorldLearning is related to work that treats large language models as sources of cultural or geographic knowledge. Recent studies probe LLMs to see how they encode cross-country variation in moral norms and values, often by comparing model responses to survey data such as the World Values Survey. For example, Ramezani and Xu show that English LLMs contain structured knowledge about moral norms in different countries, and analyze how closely these match human judgments. Tao et al. use World Values Survey questions to visualize cultural bias in several popular LLMs on the Inglehart–Welzel cultural map, highlighting a tendency toward Western, self-expression values. More recent work and surveys extend this line to broader cultural awareness and value judgments across countries and demographic groups. Within GeoAI, other systems integrate LLMs with geographic information tools to allow users to query spatial data in natural language. For instance, work on “Autonomous GIS” uses LLMs as controllers over GIS toolchains, letting users request complex spatial analyses conversationally. Commercial platforms such as ArcGIS now expose similar capabilities, positioning LLMs as front ends for map-based analysis and geoprocessing. These systems treat the model as a flexible interface over existing geospatial data and tools.

Most of this work, however, still frames interaction as question–answer: the model is asked to state what is true about a region (for example, “What do people in country X think about Y?” or “What is the land-use pattern here?”), and its text output is the final product to be evaluated or consumed. In contrast, WorldLearning uses the LLM as an approximate prior over how a shared experience (“having fun”) is realized through concrete activities, turns that prior into a set of standardized scores across all countries, and visualizes the structure of those scores so that people can inspect and critique it. The model is not treated as an oracle that tells users what is true about each country; instead, it is a generator of patterns and hypotheses that the interface exposes, and that users can question, compare, and refine through their own reasoning.

Taken together, this prior work shows that (1) experiences vary across regions, (2) maps are powerful tools for making

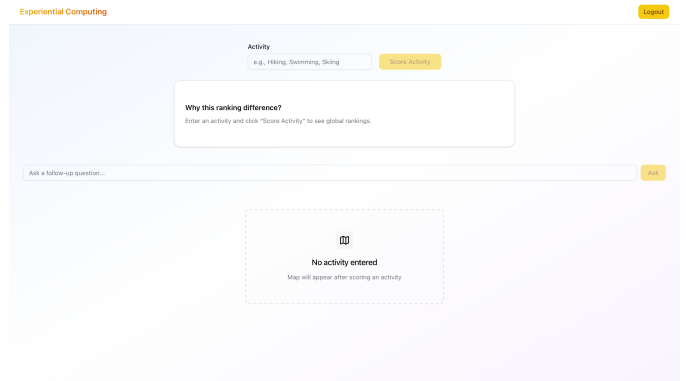


Fig. 1. Starting interface

large scale patterns visible, and (3) LLMs encode rich but uneven cultural associations. WorldLearning sits at the intersection of these threads. Its contribution is to show how a language model’s experiential associations about “what local people do for fun” can be externalized as a global pattern, embedded in an interface designed to elicit “why” questions, and used not just for comparison but also for reflecting on both difference and shared human experiences.

II. SYSTEM DESCRIPTION

A. User Interface

WorldLearning is designed for people who are curious about what local people usually do for fun in different countries, but who may not yet know what questions to ask or where to start. The system treats them as exploratory learners rather than information retrievers: instead of asking them to type a fully formed question, it invites them to propose an activity that might be “something people do for fun,” then lets them see how that activity plays out across the world. At a high level, users are allowed to enter any activity they are interested in, view a global map that shows where that activity appears more or less typical for locals, inspect specific countries in more detail, and ask follow-up questions about surprising differences the visualization reveals.

When a user opens WorldLearning, they see a simple starting view centered on a single text box labeled **Activity** and a **Score Activity** button (Figure. 1.). A short prompt card beneath it reads: “Why this ranking difference? Enter an activity and click ‘Score Activity’ to see global rankings,” and the main map area shows a placeholder message: “No activity entered. Map will appear after scoring an activity.” At this point, the user is invited to type in any activity that could be something local people do for fun in their free time (for example, “hiking” or “follow K-pop”) and press **Score Activity** to begin the exploration.

After the user submits an activity, a small confirmation banner appears at the top of the screen (for example, “Activity ‘follow Kpop’ scored successfully”), indicating that the system has run the scoring pipeline for all countries (Figure 2). The activity field remains at the top, now with a Re-score

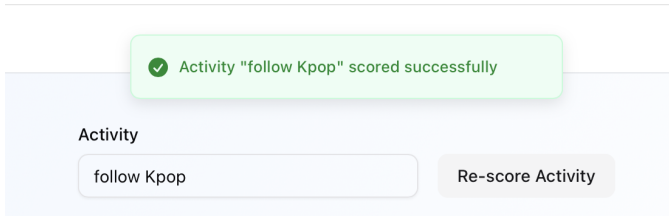


Fig. 2. Activity entered and scored

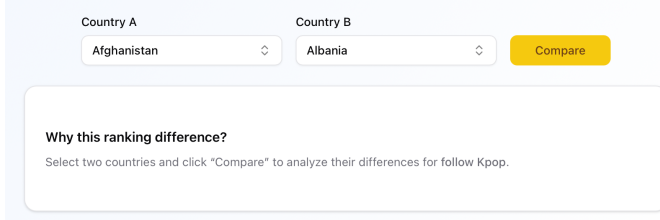


Fig. 3. Activity comparison panel

Activity button, and a comparison panel appears underneath with two dropdowns labeled Country A and Country B and a Compare button (Figure 3). A prompt card in this panel reads: “Why this ranking difference? Select two countries and click ‘Compare’ to analyze their differences for [activity],” signaling that the interface is not only about seeing scores but about understanding contrasts. Beneath this panel, the world map is displayed as a choropleth (Figure 4). Each country is shaded according to its log-probability-based score for the current activity, with a color bar labeled “[activity] Score” showing the range from “Worst” to “Best” and a caption explaining that higher scores (less negative log probabilities) correspond to more likely or typical activities. Users can pan and zoom the map, and hovering over a country reveals a tooltip with the country name, its score, and its global rank (for example, “China — follow Kpop: 18.251, Rank: #7 of 177”). Scrolling down exposes a Country Rankings section (Figure 5) that lists the Top 10 Countries and Bottom 10 Countries for the current activity, ordered by score, with a short caption reminding users that “Rankings [are] based on log-probability scores; higher scores indicate more typical activities.” Together, the map and rankings make it easy to see clusters where an activity is very typical, outliers where it is unexpectedly rare, and surprising individual countries, and in turn piques curiosity in the user and prompt them to ask more questions. A text box labeled **Ask a follow-up question...** runs across the interface above the map (Figure 6). After seeing a pattern or a surprising ranking difference, users can type a question here (for example, “Why is Greenland so low compared to South Korea for following K-pop?”). Combined with the Country A / Country B comparison controls, this field is designed to pull users into the next stage of interaction, where they move from simply viewing the global ranking to explicitly asking about specific contrasts that the visualization has made salient.

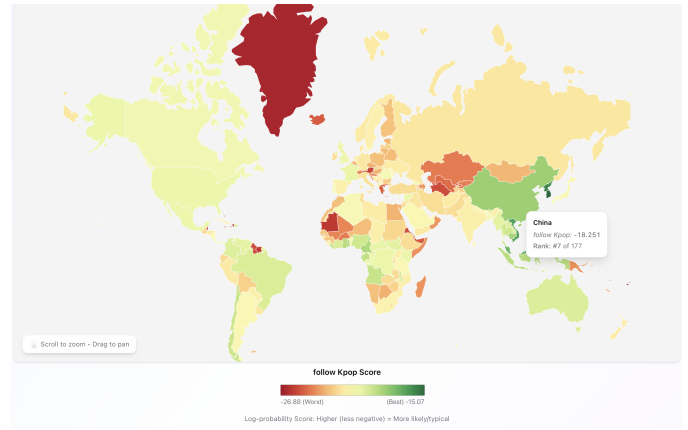


Fig. 4. Choropleth map

Country Rankings for follow Kpop

Top 10 Countries		Bottom 10 Countries	
Countries where follow Kpop is most likely/typical		Countries where follow Kpop is least likely/untypical	
1 South Korea	-15.074	168 Vanuatu	-26.879
2 North Korea	-15.867	169 Fiji	-26.853
3 Philippines	-16.323	170 Greenland	-26.477
4 Vietnam	-16.584	171 Haiti	-26.448
5 Malaysia	-17.202	172 Mauritania	-25.909
6 Taiwan	-17.212	173 Suriname	-25.791
7 China	-18.251	174 Belize	-25.745
8 Cambodia	-18.460	175 Austria	-25.724
9 Togo	-18.494	176 Guyana	-25.397
10 Cameroon	-18.668	177 Turkmenistan	-25.395

Rankings based on log-probability scores • Higher scores indicate more typical activities

Fig. 5. Top and bottom 10 ranking

B. Technical Details

The technical implementation of WorldLearning is organized around the goal of turning a large language model into an approximate “local typicality” model for leisure activities, and to expose that model in a way that supports the curiosity loop described in the introduction. This section highlights the main challenges in making that work, how the system evolved to address them, and what these design choices enable.

The first challenge was to estimate, for any activity and any country, how typical it is for local people to do that activity for fun in their free time, in a way that is comparable across the globe. WorldLearning addresses this by using davinci-002 as a scoring model. For each country–activity pair, the system constructs the same short, neutral sentence prefix: “In [country], local people usually have fun by ...”. Only the country name changes in this prefix. The activity is then appended directly after “by” (for example, “In the United States, local people usually have fun by watching football”). The model is asked

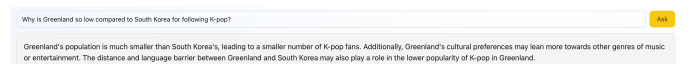


Fig. 6. Follow-up question bar

for token-level log probabilities for the entire sentence, but WorldLearning only uses the log probabilities for the activity tokens as a plausibility score. Intuitively, if the model has often seen that activity associated with that country in the context of “having fun,” it assigns higher probability to those tokens; if not, the probability is lower.

The second challenge was to represent the country scores in a form that supports global comparison without overwhelming the user or the system. When a user submits an activity, WorldLearning computes scores for all countries in one batch. From this vector, the system derives several secondary representations: a continuous color scale for the choropleth, per-country ranks, and sorted lists of the highest- and lowest-scoring countries. Early prototypes only attached scores to the map. This made it hard to tell how extreme a particular color was, or to identify specific outliers. Iterating on the design led to the current structure, in which the system (1) anchors the color scale to the global distribution of scores for the activity, and (2) exposes the tails of that distribution explicitly as “top” and “bottom” rankings. Technically, this means that each scoring run produces not just 177 independent values, but a coherent set of derived summaries (distribution statistics and order information) that are reused across the visualization. This choice matters for the paper’s hypothesis: it is what makes it possible to treat the model as a global comparator and to talk meaningfully about where an activity is unusually typical or atypical, rather than just showing a heatmap.

The third challenge was to incorporate “why” questions into the technical pipeline without hard-coding explanations or treating the language model as an oracle of facts. WorldLearning addresses this by adding two layers on top of the scores. First, when the user selects two countries for an activity, the system looks up their scores, identifies which country scores higher, and computes the size of the gap. These values are then embedded directly into a prompt that describes the situation in natural language (for example, listing both scores, noting that higher scores mean the activity is more typical, and stating which country is higher by how much). The language model is instructed to act as a cultural analyst and to explain, in a small number of sentences, why the higher-scoring country might have a higher score and what cultural, geographic, or lifestyle factors could make the activity more common there. This keeps the roles clear: the scoring pipeline defines the quantitative difference, and the explanation layer’s job is to explain that specific difference.

Second, a separate explanation endpoint handles more open-ended follow-up questions that users type after seeing the map and rankings. This endpoint accepts a natural-language query (for example, “Why might hiking be less common in this region?”) and passes it to gpt-3.5-turbo model with a fixed instruction to “explain cultural and experiential questions clearly and concisely” in a few sentences. Unlike the comparison layer, it does not reconstruct the numeric context itself; instead, it assumes the user is already looking at a pattern and wants a compact, culturally grounded explanation.

These implementation choices matter because they deter-

mine what kinds of interactions WorldLearning can support. The standardized scoring pipeline, built around a single “have fun by doing...” template and activity-token log probabilities, makes the model’s beliefs about activities comparable across all countries. The derived summaries (color scale, ranks, extremes) turn those numbers into a global structure that can be inspected from multiple angles. The contrast and explanation layers then translate specific numeric differences and user-posed “why” questions into structured prompts that elicit short, hypothesis-like explanations instead of fixed facts. Together, these technical decisions operationalize the core idea of the project: using a language model not just to answer pre-formed questions, but as an inspectable prior about everyday activities that can be probed, contrasted, and used to sustain a dialectical, curiosity-driven exploration.

These implementation choices matter because they shape the kind of interaction WorldLearning makes possible. The standardized scoring pipeline makes the model’s beliefs about activities comparable across all countries. The map and rankings turn those numbers into patterns and contrasts that are easy to see and talk about. The comparison and explanation mechanisms then turn specific differences into prompts for “why here and not there?” questions. Together, they operationalize the core idea of the project: using a language model not just as a generator of answers to well-formed questions, but as an inspectable prior about everyday activities that can be visualized, questioned, and used to sustain a curiosity-driven, back-and-forth exploration.

III. STUDY

I conducted two separate user tests with two participants. Both were university students who were familiar with web interfaces but had not been involved in building WorldLearning. Each participant used the deployed system in a one-on-one, guided session.

The study followed a simple pattern: I first gave a short explanation of what the system does at a high level (“you can enter an activity and see how typical it is as something people do for fun in different countries”), then asked the participant to use the interface while I guided them through its main capabilities. In each session, I asked them to:

- enter several activities they were curious about,
- look at the resulting world map and country rankings,
- point out anything that looked surprising or confusing,
- compare a couple of specific countries using the comparison feature, and
- ask follow-up “why” question using the explanation box.

I adapted which activities and countries we looked at based on what each participant found interesting. During the sessions, I paid attention to what they clicked on, which parts of the interface they used (map, rankings, comparison, explanations), and what kinds of comments or questions they voiced in response to what they saw.

A. Measures

Given the small sample size and the exploratory nature of the study, I focused on qualitative observations rather than quantitative. I paid attention to three aspects:

- What participants actually did with the interface I noted down how many activities they tried, whether they only checked “obvious” countries or also explored unexpected ones, and whether they used comparison and explanation features once the map appeared.
- Moments of surprise or question-generation Whenever a participant said things like “I didn’t expect that,” “why is this country so high/low?” or “now I want to know...,” I treated that as evidence that the interface had surfaced a pattern and triggered a new question.
- How they talked about the scores and explanations I noted down whether they treated the output as facts, as “what the model thinks,” or as hypotheses they could agree or disagree with, and whether this helped or hindered their exploration.

B. Results and analysis

Across the two sessions, participants engaged with the system in multiple exploration cycles: they entered more than one activity, scanned the map and rankings, and then drilled down into individual countries or pairs of countries. Neither participant stopped after a single activity; once they understood the pattern of “enter activity → see world pattern,” they started trying additional activities that reflected their own interests or intuitions about different regions.

In both sessions, participants used more than one part of the interface: they looked at the choropleth, referred to the top/bottom country rankings, and tried at least one country comparison and one free-form follow-up explanation. This suggests that, even in a small sample, the system was usable enough for people to complete the full curiosity loop it was designed for: from input, to global visualization, to contrast, to “why” questions.

RQ1: Surfacing patterns users would not have asked for

Both participants encountered patterns they described as surprising or as something they “wouldn’t think to search.” They often started with an activity they already associated with a specific place, but then noticed other countries ranked unexpectedly high or low and shifted their attention there. For example, after seeing that an activity was highly ranked in a familiar region, they scrolled the rankings and reacted to countries near the top or bottom that they had not mentioned before.

These moments are exactly what the system is designed to create: the user did not come in with a question like “How does [activity] vary between these two specific countries?” Instead, the global pattern made certain contrasts visible first, and only then did the participant form a new question. This provides initial, qualitative support for the claim that exposing the model’s associations as a world map and rankings can help users notice experiential differences they would not have thought to query directly.

RQ2: From seeing patterns to asking “why here and not there?”

Once participants noticed a surprising country or contrast, they naturally shifted into “why” mode. In both sessions, after an initial scan of the map and rankings, participants began to articulate contrastive questions in their own words, such as “why is it so high here and so low there?” They then used the comparison feature to formally compare two countries and read the resulting explanation, and in at least one case used the follow-up question box to ask an additional “why” question.

Because the sessions were guided, I did occasionally prompt them to try comparison or explanation if they had not yet done so, but I did not tell them what to ask. The content of their questions came directly from patterns they had just seen. This suggests that, once the global structure is visible, the interface can support the next step in the curiosity loop: turning an initially vague reaction (“that looks odd”) into a concrete “why here and not there?” question and giving the user tools to poke at it.

RQ3: Interpreting scores and explanations as model beliefs, not final answers

Participants did not treat the scores and explanations as unquestionable truth. Instead, they tended to interpret them as “what the model thinks” and compare that with their own intuitions or knowledge. In places where the output aligned with their expectations, they took it as confirmation that the system was capturing something reasonable. In places where it did not, they either tried to reason about why the model might see things that way or flagged it as something they would want to verify with other sources.

For explanations, participants read the generated text but also commented on where it seemed plausible, where it felt generic, and what it might be missing. This behavior is important for the project’s goals: it indicates that, at least in these small tests, the implementation encourages users to treat scores and explanations as hypotheses and prompts for further thinking, rather than as final answers that end the conversation.

IV. DISCUSSION

WorldLearning shows that we can use a large language model as an approximate “local typicality” model for everyday activities and expose its beliefs as a global pattern that people can see, question, and explore. Even in a small, guided study with two users, the system helped participants notice patterns they would not have thought to search for, and pushed them toward “why here and not there?” questions rather than stopping at a list of things to do.

Below, I highlight three design elements that made this work and that could be useful for other systems that want to support curiosity-driven exploration with LLMs.

Design lesson 1: Treat LLMs as priors over realizations of an experience

A first technique is how WorldLearning uses an LLM. Instead of asking for a direct answer like “What do people do for fun in [country]?” and then trying to compare paragraphs, the system fixes the frame and varies only the realization. The

prompt is always a short, neutral sentence about locals having fun; the only part that changes is the activity that fills that slot. The system then scores just the activity tokens.

This design does two things. Technically, it gives a comparable number for every (country, activity) pair without writing separate prompts or rules for each region. Conceptually, it reframes what the LLM is for. The model is not a travel writer that produces one answer per country; it is a prior over how a shared experience like “having fun” is realized in different places. Other systems that care about experiences rather than facts could reuse this pattern: keep the experiential frame fixed, treat the concrete realization as the variable, and use the model to score realizations instead of generating full descriptions.

Design lesson 2: Show the whole distribution and make extremes easy to see

A second technique is to always work at the level of the full distribution of scores for a given activity and to expose that distribution in multiple coordinated views. When the user scores an activity, the system computes scores for all countries in one batch. From that vector, it generates a choropleth, ranks every country, and extracts the top and bottom countries.

In the study, this combination mattered. Participants did not come in with finely targeted questions like “Compare country X and country Y for this activity.” They started with some rough expectation (for example, “I think this should be high in one region”), then scanned the map and rankings. What triggered new questions were the extremes and outliers: a country they never thought about appearing very high, or a country they expected to be high appearing low. Once those contrasts were visible, participants began to ask why.

The generalizable point is that if we want people to ask questions they did not already have, we should show them the whole shape of the model’s beliefs, not just respond to a single query. Representations that make extremes and contrasts obvious – like distributions, rankings, and explicit tails – are useful not only for analysis but also as triggers for curiosity.

Design lesson 3: Keep numbers and narrative in conversation, not collapsed

A third lesson is about how to integrate “why” with model-based visualizations. WorldLearning separates three roles. The scoring logic defines the numeric pattern. A comparison layer pulls out simple contrasts like “country A scores higher than country B by this amount”. Explanation endpoints then turn either those contrasts or free-form user questions into short, hypothesis-like narratives.

In the study, this separation seemed to help. Participants read the explanations, but they did not treat them as the final word. They compared them to what they saw on the map and in the rankings, agreed with parts, disagreed with others, and added their own reasons. The explanation text became one voice in a conversation, not the official caption of the map.

This suggests a broader design pattern for LLM-backed systems: keep numeric structure and explanatory text connected but not fused. Let the model propose possible stories grounded in visible patterns, but leave the pattern itself in front of the user. When explanations are clearly “one possible story about

what you are seeing” instead of “the answer”, people are more likely to keep thinking and asking questions, which is what systems like this are trying to support.

A. Limitations

This work has several important limitations. The study is very small: two guided sessions cannot tell us how a broader population would use or misunderstand the system. Participants were also students who are relatively comfortable with data and AI tools, people with different backgrounds might interpret the scores and explanations differently.

On the modeling side, WorldLearning directly reflects the biases and blind spots of the underlying language models. The system presents a single score per country for each activity and does not expose uncertainty or disagreement between prompts or models. In the study, participants sometimes noticed when an output felt stereotypical or implausible, but the interface does not actively help them detect or organize such concerns.

B. Future work

This paper focuses on a single underlying experience, “having fun in free time,” but the same approach could be extended to a broader set of experiences. Future work could include experiences like relaxing, socializing or commuting, and examine how different categories of experience generate different kinds of patterns and questions. Comparing, for example, “how people relax” versus “how people socialize” across the same set of countries could reveal whether the same places that cluster together for fun also cluster for other aspects of everyday life.

On the modeling side, WorldLearning currently uses one model and one prompting strategy. An obvious next step is to explore other models and configurations, such as newer LLMs, and to systematically compare their scoring results. This could help distinguish which patterns are robust across models. It would also open up the possibility of letting users directly compare “what this model believes” with “what that model believes” about the same activity.

The explanation layer also has room to grow. The current implementation returns short, single-paragraph explanations. Future work could experiment with richer, more structured explanations that explicitly separate different types of drivers (for example, climate, infrastructure, media, policy), or include pictures in explanations. Another direction is to give users more control over explanation style, such as asking for more speculative versus more conservative answers, or requesting alternative explanations when they disagree with the first one.

Finally, the current interface primarily shows a global view. Many of the questions that arose in the study, however, pointed toward finer-grained variation within a single country, such as urban versus rural differences or regional subcultures. Future work could “zoom in” on one country by breaking the map into subregions. This would preserve the global comparison that makes patterns visible, while allowing more detailed exploration of what “local people” means in any given place.

Taken together, these directions point to a broader goal that goes beyond comparison alone. By mapping how shared experiences are realized differently and similarly around the world, systems like WorldLearning could become tools for exploring our shared humanity: not just where we differ, but how often we are trying to do the same things with the lives we have, under very different conditions.

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